MIL-C-915/54B
5 April 1973
SUPERSEDING 1/
MIL-C-915/54A
1 August 1972

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL 600 VOLTS, TYPE ECM

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the cable described herein shall consist of this document and the latest issue of Specification MIL-C-915.

REQUIREMENTS:

Qualification required.

Construction (Watertight)

Twisted Pairs (8 pairs):

First - Copper conductor, ASTM B-286 size 20-7, tin coated. Maximum diameter 0.039 inch.

Second - Polyethylene insulation, type II, class L, grade 4 of L-P-390. Nominal thickness 0.017 inch. Colored insulation - one black and one white conductor for each pair.

Third - Clear polyamide jacket on each conductor, type III, grade E or type IV of MIL-M-20693. Nominal thickness 0.003 inch.

Fourth - Two conductors (one black and one white) cabled together with a nominal lay of 2.5 inches, to form a pair.

Fifth - Binder tape over each pair, at manufacturer's option, applied helically with overlap.

Sixth - Braided shield of AWG No. 34 or 36 tin coated copper. Braid angle of 30 to 35 degrees; minimum coverage of 85 percent.

Seventh - Shield insulation of one polyester tape, type G of MIL-I-631, plus a jacket of clear polyamide, type III, grade E or type IV of MIL-M-20693. Nominal thickness 0.003 inch. Standard Identification Code, applied by method 2.

- Alternate shield insulation of two polyester tapes, type G of MIL-I-631, sealed. Standard Identification Code, applied by method 2, on inner tape.

Groups of 7 (8 groups):

(B) First - Copper conductor, AWG No. 18 (7 X 0.0152 inch), tin coated. Maximum diameter 0.050 inch.

Second - Polyvinyl chloride insulation (80°C.). Nominal thickness 0.008 inch. Colored insulation - 5 black and 2 white conductors for each group.

Third - Seven conductors (5 black and 2 white, with the 2 white conductors adjacent in the layer of six) cabled with a lay not greater than 2 inches, to form each group.

Fourth - One polyester tape, type G of MIL-I-631, plus a jacket of clear polyamide, type III, grade E or type IV of MIL-M-20693. Nominal thickness 0.003 inch. Standard identification code, applied by method 2. Alternate of two polyester tapes, type G of MIL-I-631, sealed. Standard identification code, applied by method 2, on inner tape.

ASSEMBLY:

First - Five pairs cabled to form a core. Three pairs and eight groups of seven cabled over the core, with a lay not greater than 24 times the pitch diameter of the layer. Pair Nos. 1 through 5 in core; the sequence of the outer-layer - pair No. 6, group Nos. 1, 2 and 3; pair No. 7; group Nos. 4, 5 and 6; pair No. 8; group Nos. 7 and 8. Non-fibrous fillers shall be employed to form a well-rounded assembly.

1/ Requirements formerly covered on MIL-C-24145/14.

(B) Denotes changes

FSC 6145

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T12:12Z
Check the source to verify that this is the current version before use.

ASSEMBLY (Cont'd):
Second - Binder tape applied helically with overlap.
Third - Standard thermoplastic jacket (gray). Nominal thickness 0.100 inch. Cable surface marking required.

NOTE: Completed cable diameters: Minimum 1.290 inches, maximum 1.370 inches.

EXAMINATION AND TESTS:

	Requirements:
Basic electrical:	
Conductor resistance - At 25°C., ohms/1000 ft., maximum	
(B) ASTM B-286 size 20-7.	11.03
(B) AWG No. 18	7.35
Voltage withstand - Volts, rms, minimum	
Conductor to conductor.	2000
Conductor to shield	1000
Shield to shield.	500
Insulation resistance (polyvinyl chloride only) -	
Megohms/1000 ft., minimum	500
Group A:	
Visual and dimensional	
Watertightness - Leakage at 25 psi, 6 hrs., cu. in., max.	1.0
Capacitance	
Mutual capacitance - At 1 kHz, pF/ft. max.	30
Group B:	
Drip - At 75° ± 1°C.	Zero
Physicals (unaged)	
Insulation (conductor)	
Polyvinyl chloride	
Tensile strength - psi, minimum	1800
Elongation - Percent, minimum	100
Polyethylene	
Tensile strength - psi, minimum	1400
Elongation - Percent, minimum	300
Jacket (cable)	
Tensile strength - psi, minimum	1800
Elongation - Percent, minimum	225
Group C:	
Physicals (aged)	
Insulation (conductor)	
Air oven (polyvinyl chloride)	
Elongation - Percent of unaged, minimum	90
Air oven (polyethylene)	
Elongation - Percent of unaged, minimum	60
Jacket (cable)	
Hot oil immersion	
Tensile strength - psi, minimum	1400
Elongation - Percent, minimum	50
Permanence of printing (jacket) - Cycles, minimum.	250
Cable filler removability	

QUALIFICATION INSPECTION

Qualification inspection shall include basic electrical, all of Groups A, B and C plus the following:

Cable aging and compatibility (95° ± 3°C.)	No failure.

UNIT ORDERING LENGTHS: 1000 feet (nominal)

Custodians:
Army - MI
Navy - SH
Air Force - 80

Review activities:
Army - EL, MI, AV
Navy - EC
Air Force - 80

User activities:
Army - ME, MU, WC
Navy - CG

Preparing activity:
Navy - SH
(Project 6145-0648-15)

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T12:12Z
Check the source to verify that this is the current version before use.